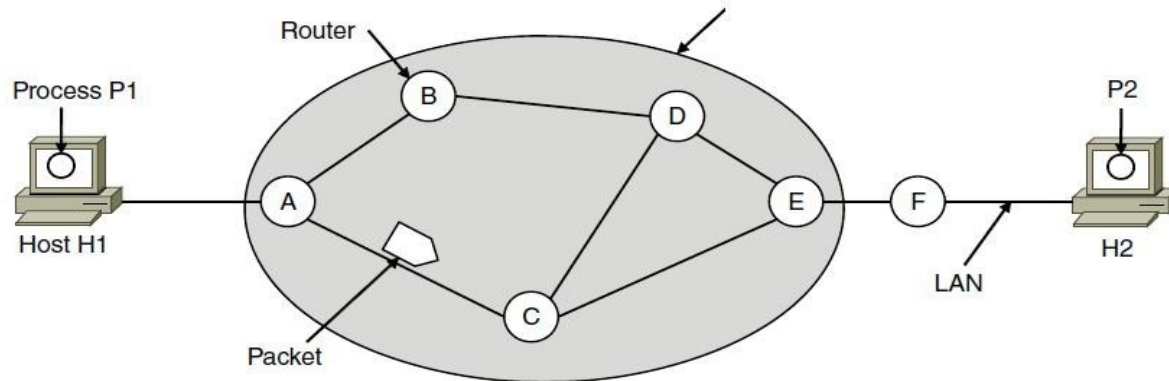


UNIT-III

Network Layer Design Issues

- Store-and-forward packet switching
- Services provided to transport layer
- Implementation of connectionless service
- Implementation of connection-oriented service
- Comparison of virtual-circuit and datagram networks

- **Store-and-forward packet switching** : A host with a packet to send transmits it to the nearest router, either on its own LAN or over a point-to-point link to the ISP.
- The packet is stored there until it has fully arrived and the link has finished its processing by verifying the checksum.
- Then it is forwarded to the next router along the path until it reaches the destination host, where it is delivered. This mechanism is store-and-forward packet switching.



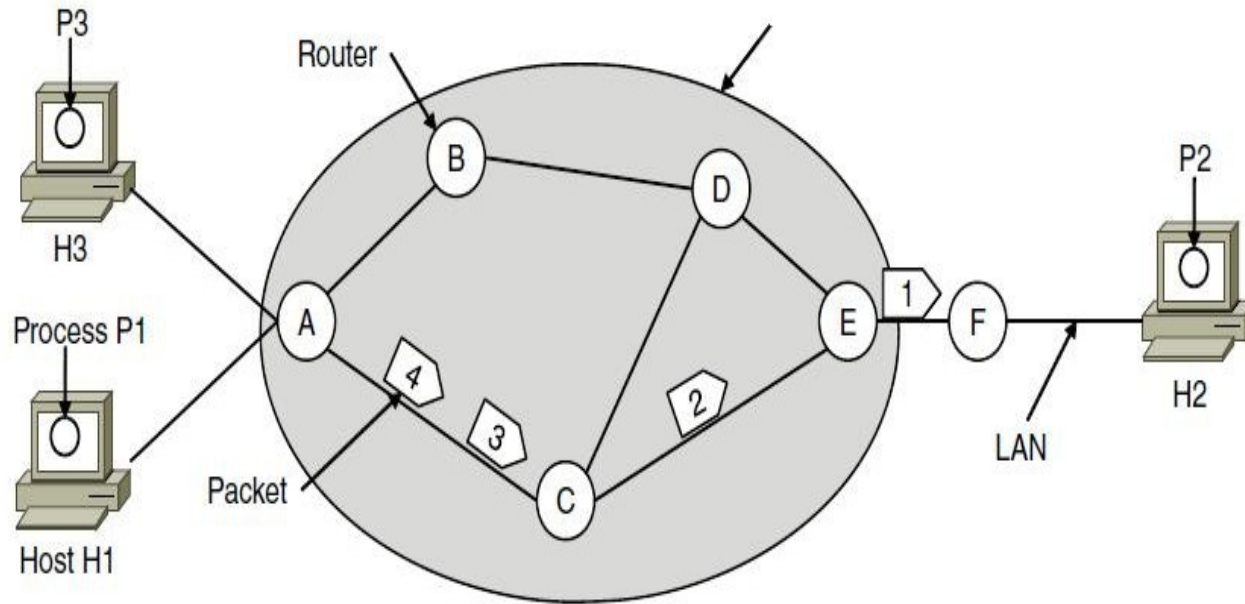
- **Services provided to transport layer**
 - The network layer provides services to the transport layer at the network layer/transport layer interface. The services
 - need to be carefully designed with the following goals in mind:
 - Services independent of router technology.
 - Transport layer shielded from number, type, topology of routers.
 - Network addresses available to transport layer use uniform numbering plan
- even across LANs and WANs

- **Implementation of connectionless service**
- If connectionless service is offered, packets are injected into the network individually and routed independently of each other.
- No advance setup is needed. In this context, the packets are frequently called **datagrams** (in analogy with telegrams) and the network is called a **datagram network**.

- Let us assume for this example that the message is four times longer than the maximum packet size, so the network layer has to break it into four packets, 1, 2, 3, and 4, and send each of them in turn to router *A*.
- Every router has an internal table telling it where to send packets for each of the possible
- destinations. Each table entry is a pair(destination and the outgoing line). Only directly connected lines can be used.
- *A*'s initial routing table is shown in the figure under the label “initially.”
- At *A*, packets 1, 2, and 3 are stored briefly, having arrived on the incoming link. Then each packet is forwarded according to *A*'s table, onto the outgoing link to *C* within a new frame. Packet 1 is then forwarded to *E* and then to *F*.

- However, something different happens to packet 4. When it gets to *A* it is sent to router *B*, even though it is also destined for *F*.
- For some reason (traffic jam along ACE path), *A* decided to send packet 4 via a different route than that of the first three packets.
- Router A updated its routing table, as shown under the label “later.”
- The algorithm that manages the tables and makes the routing decisions is called the **routing algorithm**.

- Implementation of connection-oriented service



A's table

H1	1	C	1
H3	1	C	2
In		Out	

C's Table

A	1	E	1
A	2	E	2

E's Table

C	1	F	1
C	2	F	2

- If connection-oriented service is used, a path from the source router all the way to the destination router must be established before any data packets can be sent. This connection is called a **VC (virtual circuit)**, and the network is called a **virtual-circuit network**
- When a connection is established, a route from the source machine to the destination machine is chosen as part of the connection setup and stored in tables inside the routers.
- That route is used for all traffic flowing over the connection, exactly the same way that the telephone system works.
- When the connection is released, the virtual circuit is also terminated. With connection-oriented service, each packet carries an identifier telling which virtual circuit it belongs to.

- This leads to the second row in the tables. Note that we have a conflict here because although A can easily distinguish connection 1 packets from $H1$ from connection 1 packets from $H3$, C cannot do this.
- For this reason, A assigns a different connection identifier to the outgoing traffic for the second connection.
- Avoiding conflicts of this kind is why routers need the ability to replace connection identifiers in outgoing packets.
- In some contexts, this process is called **label switching**.
- An example of a connection-oriented network service is **MPLS (Multi Protocol Label Switching)**.

- **Comparison of virtual-circuit and datagram networks**

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

- **Routing Algorithms**
- The main function of NL (Network Layer) is routing packets from the source machine to the destination machine.
- There are two processes inside router:
- One of them handles each packet as it arrives, looking up the outgoing line to use for it in the routing table. This process is forwarding.
- The other process is responsible for filling in and updating the routing tables. That is where the routing algorithm comes into play. This process is routing.

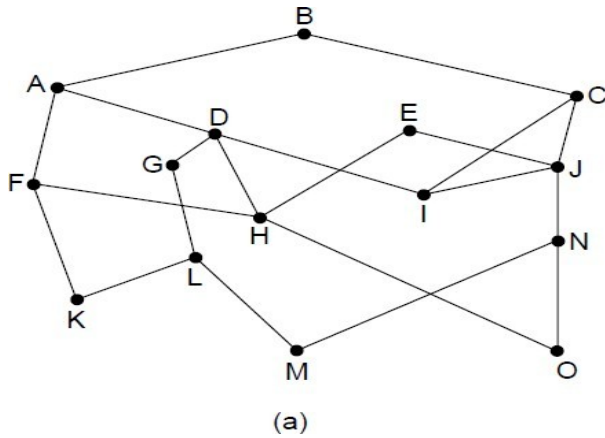
- Regardless of whether routes are chosen independently for each packet or only when new connections are established, certain properties are desirable in a routing algorithm **correctness, simplicity, robustness, stability, fairness, optimality.**
- Routing algorithms can be grouped into two major classes:
 - nonadaptive (Static Routing)
 - adaptive. (Dynamic Routing)
- Nonadaptive algorithm do not base their routing decisions on measurements or estimates of the current traffic and topology. Instead, the choice of the route to use to get from I to J is computed in advance, off line, and downloaded to the routers when the network is booted. This procedure is sometimes called static routing.
- Adaptive algorithm, in contrast, change their routing decisions to reflect changes in the topology, and usually the traffic as well.

Different Routing Algorithms

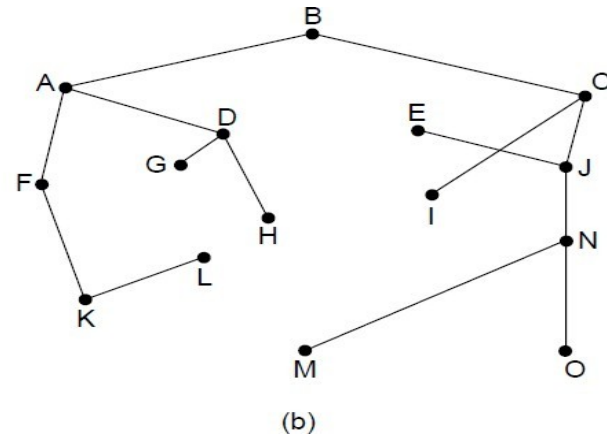
- Optimality principle
- Shortest path algorithm
- Flooding
- Distance vector routing
- Link state routing
- Hierarchical Routing

- **The Optimality Principle**
- One can make a general statement about optimal routes without regard to network topology or traffic. This statement is known as the optimality principle.
- It states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same
- As a direct consequence of the optimality principle, we can see that the set of optimal routes from all sources to a given destination form a tree rooted at the destination. Such a tree is called a **sink tree**.
- The goal of all routing algorithms is to discover and use the sink trees for all routers.

The Optimality Principle



(a) A network.



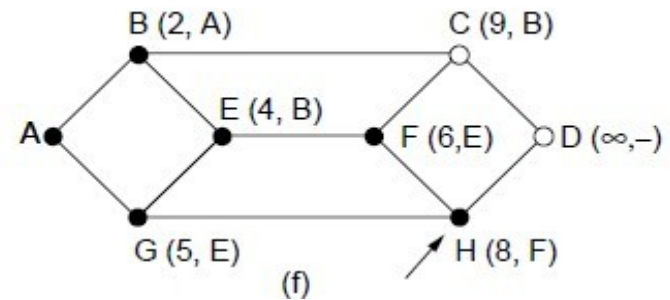
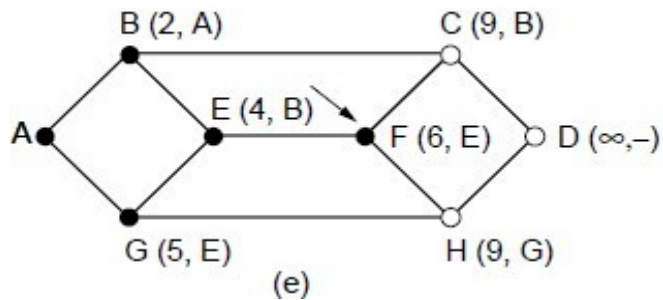
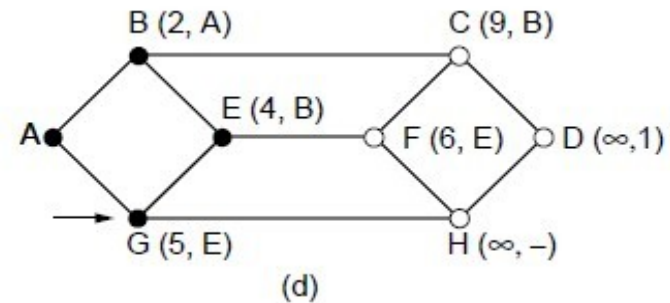
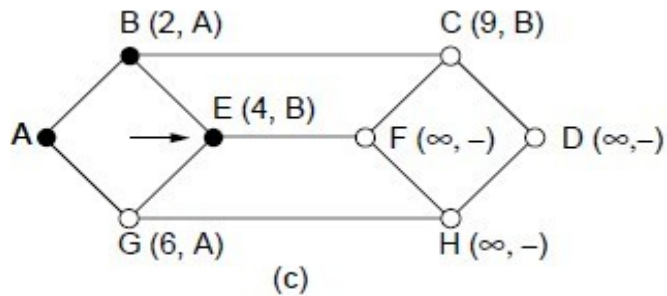
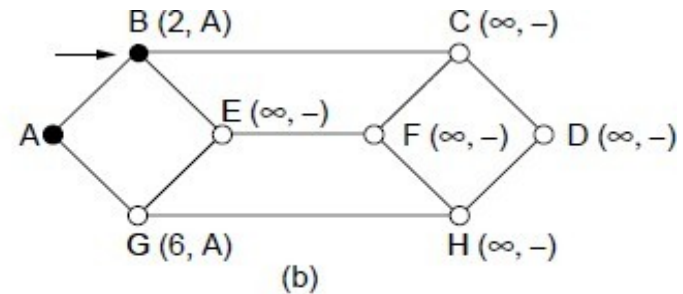
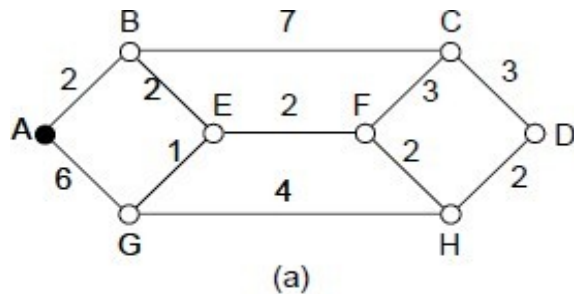
(b) A sink tree for router *B*.

- **Shortest Path Routing (Dijkstra's)**
- The idea is to build a graph of the subnet, with each node of the graph representing a router and each arc of the graph representing a communication line or link.
- To choose a route between a given pair of routers, the algorithm just finds the shortest path between them on the graph
- Start with the local node (router) as the root of the tree. Assign a cost of 0 to this node and make it the first permanent node.

- Examine each neighbor of the node that was the last permanent node.
- Assign a cumulative cost to each node and make it tentative
- Among the list of tentative nodes
 - Find the node with the smallest cost and make it Permanent
 - If a node can be reached from more than one route then select the route with the shortest cumulative cost.
- Repeat steps 2 to 4 until every node becomes permanent.

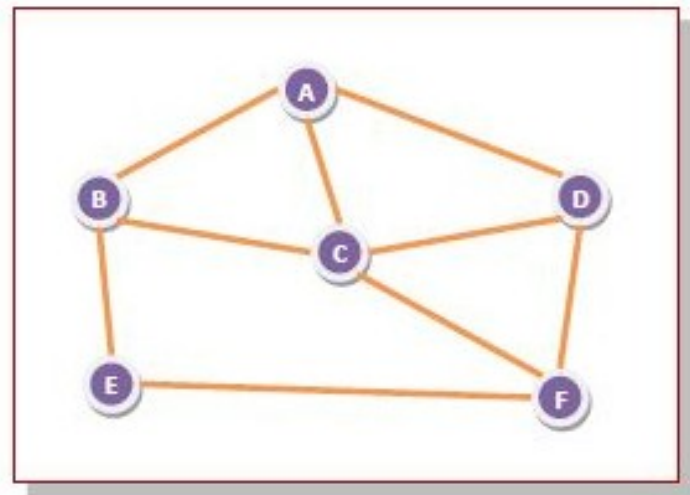
- **Key Concepts:**
- **Graph Representation:**
- The network is modeled as a graph where routers are represented as nodes (vertices) and communication links are represented as edges. Each edge has a weight associated with it, representing the cost of traversing that link.
- **Minimizing Path Cost:**
- The goal is to find a path between two nodes where the sum of the weights of the edges along that path is minimized.
- **Dynamic vs. Static Routing:**
- Shortest path algorithms can be used in both static routing (where paths are pre-determined) and dynamic routing (where paths adapt to network changes).

- Shortest Path Routing



- **Flooding**
- flooding is a simple routing technique where a sender transmits a data packet to every node in a network, sending it out through every outgoing link except the one it arrived on.
- This non-adaptive method ensures a message reaches all parts of a network, though it can generate many duplicate packets, requiring mechanisms like hop counts or sequence numbers to prevent infinite loops and wasted resources.
- Flooding is used in routing protocols, peer-to-peer file sharing, and Usenet to guarantee reachability, though it can also be exploited in denial-of-service attacks.

- **Flooding ex:**
- Using flooding technique
- An incoming packet to A, will be sent to B, C and D.
- B will send the packet to C and E.
- C will send the packet to B, D and F.
- D will send the packet to C and F.
- E will send the packet to F.
- F will send the packet to C and E.



- **Key Points:**
- **No routing table needed:** Flooding does not require any knowledge of network topology or routing tables.
- **Guaranteed delivery:** If the network is connected and no limits are imposed, flooding guarantees the packet reaches all nodes.
- **High redundancy:** Multiple copies of the same packet can travel along different paths, causing a lot of duplicates.
- **Possible problems:**
 - **Broadcast storm:** Because every node forwards packets to all neighbors, flooding can cause excessive traffic and congestion.
 - **Loops:** Packets can circulate indefinitely unless there is a mechanism to prevent this (like hop count or sequence numbers).

- Uses
- **Routing Protocol Updates:** Distributing protocol updates to all nodes in a large network.
- **Path Discovery:** Finding paths in routing protocols.
- **Bridging:** Used in some bridging algorithms.
- **P2P Systems & Usenet:** Common in peer-to-peer file sharing and Usenet networks for message distribution.
- **Wireless Ad-hoc Networks (WANETs):** Incorporated into some routing protocols for wireless networks.

- Controls to Prevent Duplicates
- Sequence Number Controlled Flooding(SNCF):
- Nodes store a memory of packet addresses and sequence numbers; if a packet with a previously seen sequence number arrives, it's dropped.
- Reverse Path Forwarding(RPF):
- A node only forwards a packet if it arrived from the correct next node in the path back to the source.
- Hop Counter:
- Each packet is given a limit on the number of hops it can travel; when the counter reaches zero, the packet is discarded.

- **Types of Flooding**

- **Uncontrolled Flooding**

The simplest form, where packets are sent to all neighbors without any conditions, potentially leading to infinite loops or "broadcast storms".

- **Controlled Flooding**

More sophisticated, using mechanisms to prevent infinite loops.

- **Hop Count (Time-To-Live or TTL):** The packet is discarded after a certain number of hops to limit its propagation.

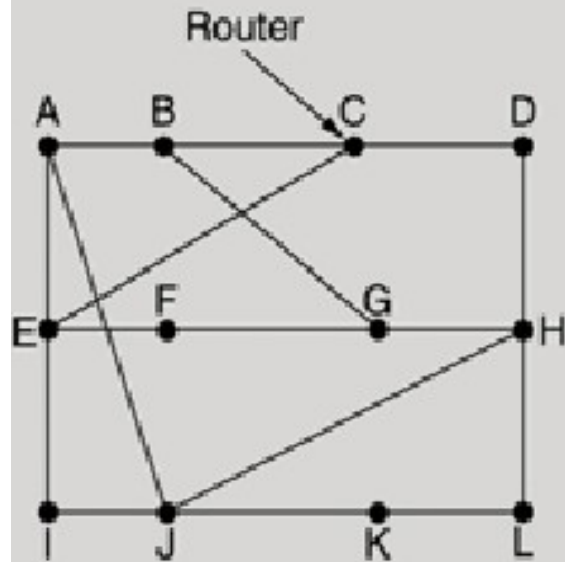
- **Sequence Numbers:** Each packet is tagged with a unique sequence number; a node will only forward a packet if it hasn't seen it before, preventing duplicates.

- **Selective Flooding:**

A form of controlled flooding that aims to flood only in specific directions, such as using Reverse Path Forwarding (RPF) to send a packet back only if it was received from the correct path.

- **Distance vector routing algorithms**

- Distance vector routing algorithms operate by having each router maintain a table (i.e, a vector) giving the best known distance to each destination and which line to use to get there.
- These tables are updated by exchanging information with the neighbors.
- The distance vector routing algorithm is sometimes called by other names, most commonly the distributed Bellman-Ford routing algorithm and the Ford-Fulkerson algorithm, after the researchers who developed it (Bellman, 1957; and Ford and Fulkerson, 1962).
- It was the original ARPANET routing algorithm and was also used in the Internet under the name RIP.



(a)

					New estimated delay from J	
					↓	Line
To	A	I	H	K		
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	—
K	24	22	22	0	6	K
L	29	33	9	9	15	K

JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6
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Vectors received from J's four neighbors

New routing table for J

(b)

The Count-to-Infinity Problem

- Failures can cause DV to “count to infinity” while seeking a path to an unreachable node

A	B	C	D	E	
•	•	•	•	•	Initially
	1	•	•	•	After 1 exchange
	1	2	•	•	After 2 exchanges
	1	2	3	•	After 3 exchanges
	1	2	3	4	After 4 exchanges

Good news of a path
to A spreads quickly

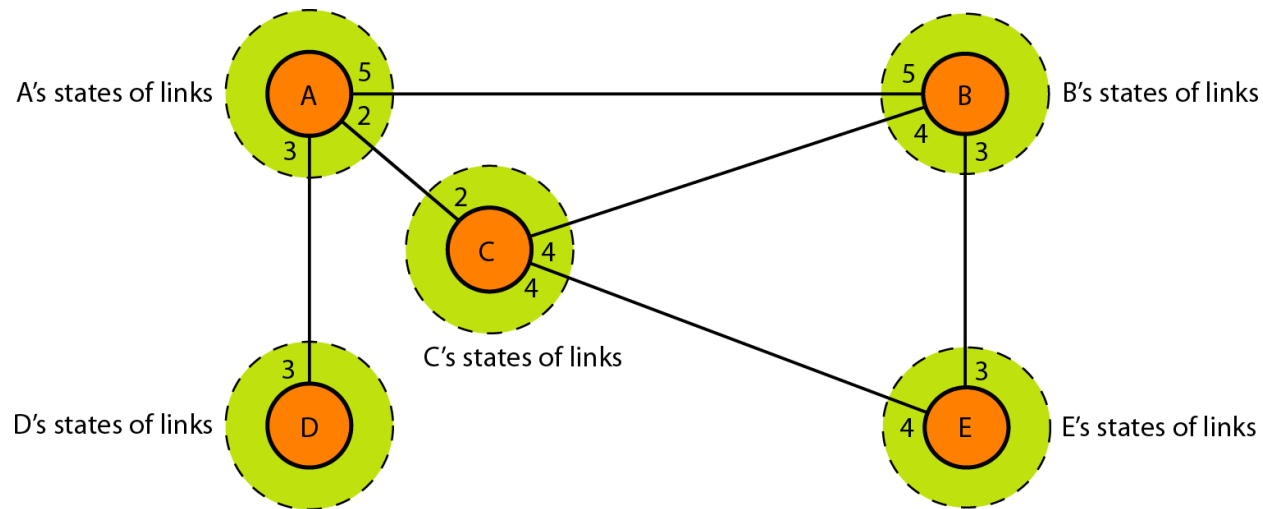
A	B	C	D	E	
•	✗	•	•	•	Initially
	1	2	3	4	After 1 exchange
	3	2	3	4	After 2 exchanges
	3	4	3	4	After 3 exchanges
	5	4	5	4	After 4 exchanges
	5	6	5	6	After 5 exchanges
	7	6	7	6	After 6 exchanges
	7	8	7	8	After 7 exchanges
	•	•	•	•	•

Bad news of *no path* to A
is learned slowly

- The main problem faced by distance vector routing is the count-to-infinity problem. Distance vector routing works in theory but has a serious drawback in practice. Although it converges to the correct answer, it may do so slowly.
- In particular it reacts rapidly to good news, but leisurely to bad news. To see how fast the good news propagates, consider the five-node subnet.
- From this figure, it should be clear why bad news travels slowly, no router ever has a value more than one higher than the minimum of all its neighbours.
- Gradually, all the routers work their way up to infinity, but the number of exchanges required depends on the numerical value used for infinity.
- For this reason, it is wise to set infinity to the longest path plus 1. if the metric is time delay, there is no well-defined upper bound, so a high value is needed to prevent a path with a long delay from being treated as down. Not entirely surprisingly, this problem is known as count to infinity problem. The solution to count to infinity problem is the, split horizon hack.

- **Link State Routing**

- Link state routing is based on the assumption that, although the global knowledge about the topology is not clear, each node has partial knowledge: it knows the state (type, condition, and cost) of its links. **In other words, the whole topology can be compiled from the partial knowledge of each node**



Building Routing Tables

- Creation of the states of the links by each node, called the link state packet (LSP).
- Dissemination of LSPs to every other router, called **flooding, in an efficient and** reliable way.
- Formation of a shortest path tree for each node.

1. Creation of Link State Packet (LSP) A link state packet can carry a large amount of information. For the moment, we assume that it carries a minimum amount of data: the node identity, the list of links, a sequence number, and age.

- The first two, node identity and the list of links, are needed to make the topology. The third, sequence number, facilitates flooding and distinguishes new LSPs from old ones. The fourth, age, prevents old LSPs from remaining in the domain for a long time.
- LSPs are generated on two occasions:
 - When there is a change in the topology of the domain
 - on a periodic basis: The period in this case is much longer compared to distance vector. The timer set for periodic dissemination is normally in the range of **60 min or 2 h** based on the implementation. A longer period ensures that flooding does not create too much traffic on the network.

2.Flooding of LSPs: After a node has prepared an LSP, it must be disseminated to all other nodes, not only to its neighbors. The process is called flooding and based on the following

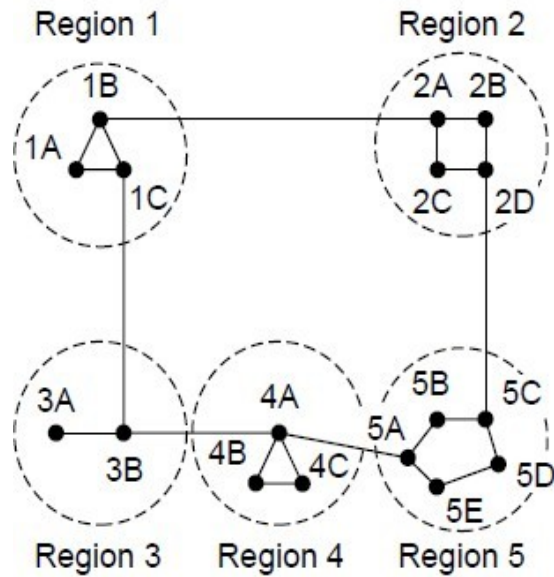
- The creating node sends a copy of the LSP out of each interface

- A node that receives an LSP compares it with the copy it may already have. If the newly arrived LSP is older than the one it has (found by checking the sequence number), it discards the LSP. If it is newer, the node does the following:
 - It discards the old LSP and keeps the new one.
 - It sends a copy of it out of each interface except the one from which the packet arrived. This guarantees that flooding stops somewhere in the domain (where a node has only one interface).
- **3.Formation of Shortest Path Tree: Dijkstra Algorithm**
 - A shortest path tree is a tree in which the path between the root and every other node is the shortest.
 - The Dijkstra algorithm creates a shortest path tree from a graph. The algorithm divides the nodes into two sets: **tentative and permanent**. It finds the neighbors of a current node, makes them tentative, examines them, and if they pass the criteria, makes them permanent.

- **Hierarchical Routing**

- As networks grow in size, the router routing tables grow proportionally. Not only is router memory consumed by ever-increasing tables, but more CPU time is needed to scan them and more bandwidth is needed to send status reports about them.
- At a certain point, the network may grow to the point where it is no longer feasible for every router to have an entry for every other router, so the routing will have to be done hierarchically, as it is in the telephone network.
- When hierarchical routing is used, the routers are divided into what we will call regions.
- Each router knows all the details about how to route packets to destinations within its own region but knows nothing about the internal structure of other regions.

- Hierarchical Routing**



(a)

Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

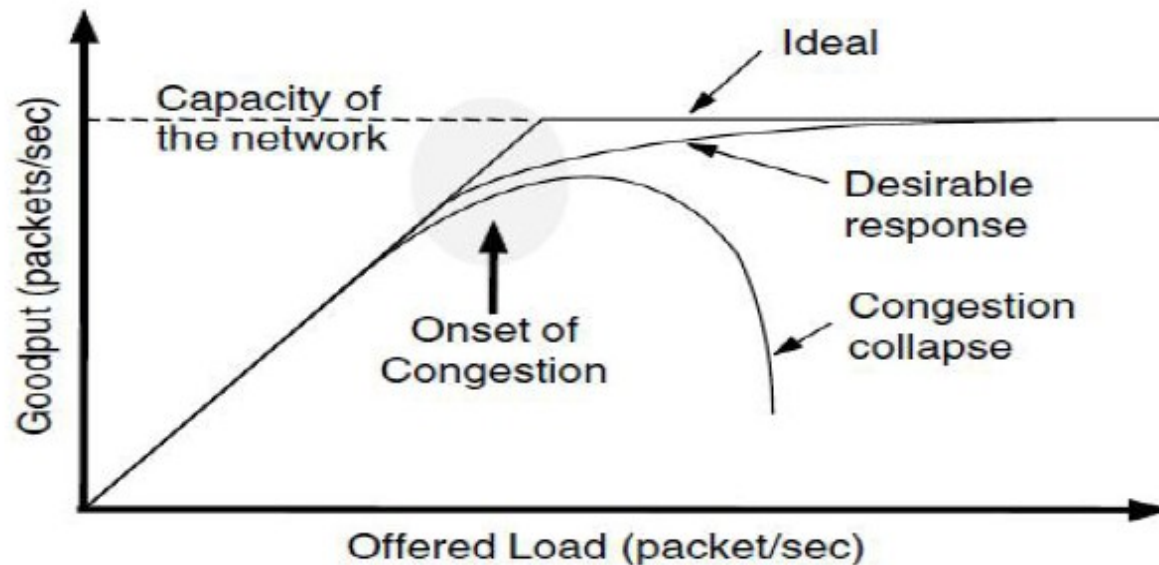
Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

- For huge networks, a two-level hierarchy may be insufficient; it may be necessary to group the regions into clusters, the clusters into zones, the zones into groups, and so on, until we run out of names for aggregations. When a single network becomes very large, an interesting question is “how many levels should the hierarchy have?”
- For example, consider a network with 720 routers. If there is no hierarchy, each router needs 720 routing table entries.
- If the network is partitioned into 24 regions of 30 routers each, each router needs 30 local entries plus 23 remote entries for a total of 53 entries.
- If a three-level hierarchy is chosen, with 8 clusters each containing 9 regions of 10 routers, each router needs 10 entries for local routers, 8 entries for routing to other regions within its own cluster, and 7 entries for distant clusters, for a total of 25 entries.

- **CONGESTION**

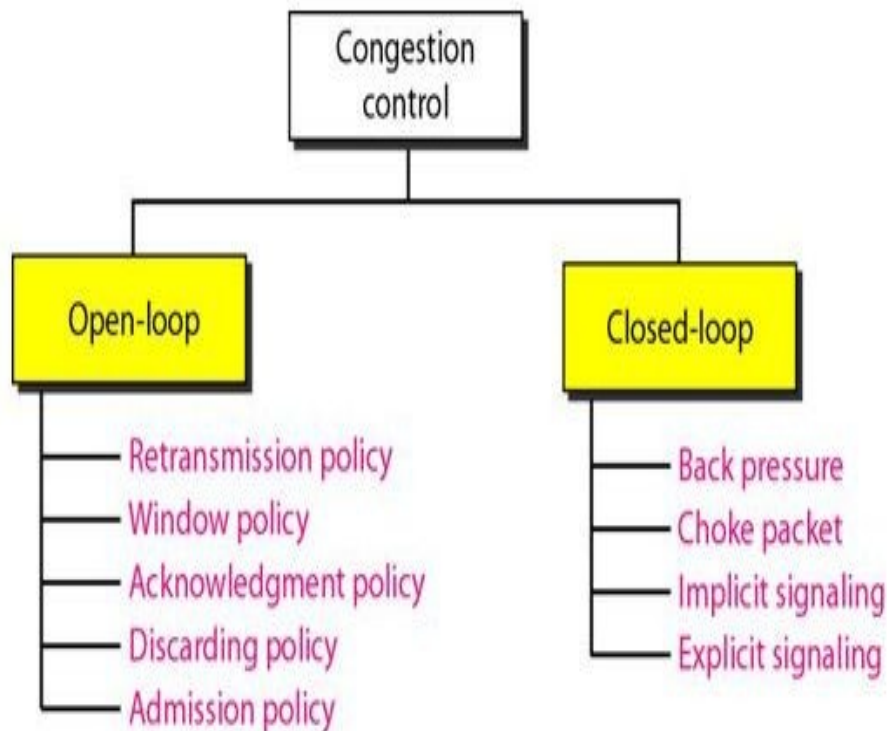
- Congestion occurs when the number of packets being transmitted through the network approaches the packet handling capacity of the network
- Congestion control aims to keep number of packets below level at which performance falls off dramatically.



- Above Figure depicts the onset of congestion. When the number of packets hosts send into the network is well within its carrying capacity, the number delivered is proportional to the number sent.
- If twice as many are sent, twice as many are delivered. However, as the offered load approaches the carrying capacity, bursts of traffic occasionally fill up the buffers inside routers and some packets are lost.
- These lost packets consume some of the capacity, so the number of delivered packets falls below the ideal curve. The network is now congested.
- Unless the network is well designed, it may experience a **congestion collapse**

What is **congestion**?

- A state occurring in network layer when the message traffic is so heavy that it slows down network response time.
- **Effects** of Congestion
 - As delay increases, performance decreases.
 - If delay increases, retransmission occurs, making situation worse.



- **Open Loop Congestion Control**

Open loop congestion control policies are applied to prevent congestion before it happens. The congestion control is handled either by the source or the destination.

- **Policies adopted by open loop congestion control –**

- **Retransmission Policy :**

It is the policy in which retransmission of the packets are taken care. If the sender feels that a sent packet is lost or corrupted, the packet needs to be retransmitted. This transmission may increase the congestion in the network.

To prevent congestion, retransmission timers must be designed to prevent congestion and also able to optimize efficiency.

- **Window Policy :**

The type of window at the sender side may also affect the congestion. Several packets in the Go-back-n window are resent, although some packets may be received successfully at the receiver side. This duplication may increase the congestion in the network and making it worse.

Therefore, Selective repeat window should be adopted as it sends the specific packet that may have been lost.

- **Discarding Policy :**

A good discarding policy adopted by the routers is that the routers may prevent congestion and at the same time partially discards the corrupted or less sensitive package and also able to maintain the quality of a message. In case of audio file transmission, routers can discard less sensitive packets to prevent congestion and also maintain the quality of the audio file

- **Acknowledgment Policy :**

Since acknowledgement are also the part of the load in network, the acknowledgment policy imposed by the receiver may also affect congestion. Several approaches can be used to prevent congestion related to acknowledgment.

The receiver should send acknowledgement for N packets rather than sending acknowledgement for a single packet. The receiver should send a acknowledgment only if it has to sent a packet or a timer expires.

- **Admission Policy :**

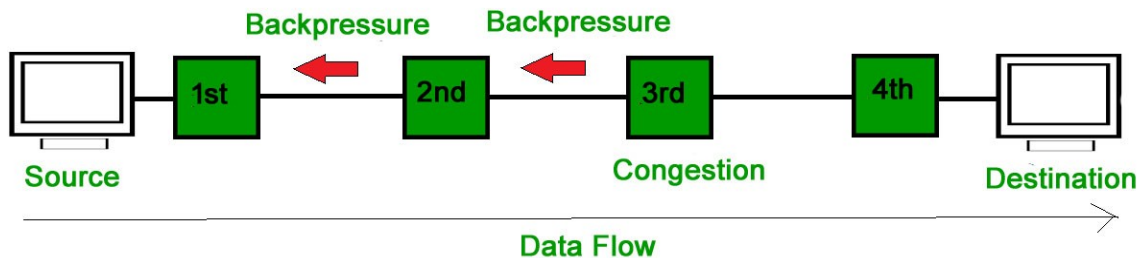
In admission policy a mechanism should be used to prevent congestion. Switches in a flow should first check the resource requirement of a network flow before transmitting it further. If there is a chance of a congestion or there is a congestion in the network, router should deny establishing a virtual network connection to prevent further congestion.

- **Closed Loop Congestion Control**

- Closed loop congestion control technique is used to treat or alleviate congestion after it happens. Several techniques are used by different protocols; some of them are:

- **1.Backpressure :**

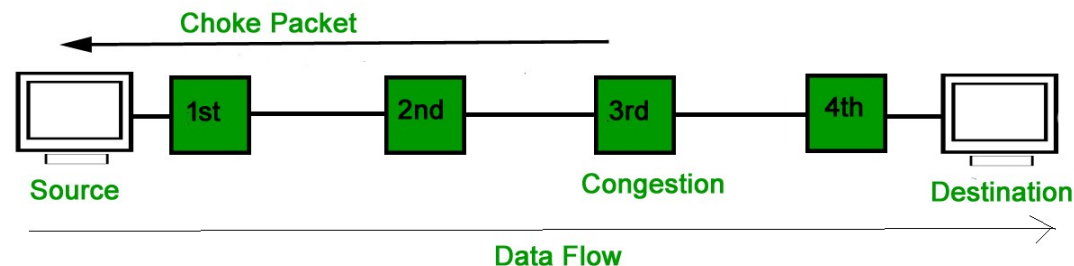
Backpressure is a technique in which a congested node stop receiving packet from upstream node. This may cause the upstream node or nodes to become congested and rejects receiving data from above nodes. Backpressure is a node-to-node congestion control technique that propagate in the opposite direction of data flow. The backpressure technique can be applied only to virtual circuit where each node has information of its above upstream node.



- In above diagram the 3rd node is congested and stops receiving packets as a result 2nd node may be get congested due to slowing down of the output data flow. Similarly 1st node may get congested and informs the source to slow down.

- **Choke Packet Technique :**

Choke packet technique is applicable to both virtual networks as well as datagram subnets. A choke packet is a packet sent by a node to the source to inform it of congestion. Each router monitor its resources and the utilization at each of its output lines. whenever the resource utilization exceeds the threshold value which is set by the administrator, the router directly sends a choke packet to the source giving it a feedback to reduce the traffic. The intermediate nodes through which the packets has traveled are not warned about congestion.



3. Implicit Signaling :

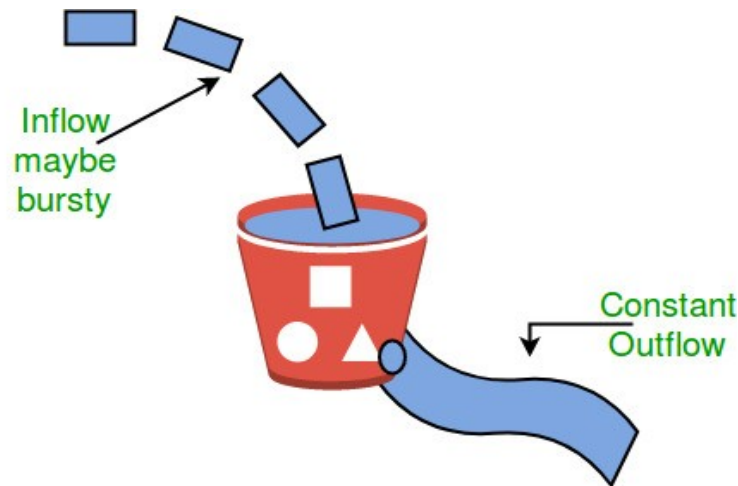
In implicit signaling, there is no communication between the congested nodes and the source. The source guesses that there is congestion in a network. For example when sender sends several packets and there is no acknowledgment for a while, one assumption is that there is a congestion.

- **3. Explicit Signaling :**

In explicit signaling, if a node experiences congestion it can explicitly send a packet to the source or destination to inform about congestion. The difference between choke packet and explicit signaling is that the signal is included in the packets that carry data rather than creating a different packet as in the case of the choke packet technique.

- Explicit signaling can occur in either forward or backward direction.
 - **Forward Signaling :** In forward signaling, the signal is sent in the direction of the congestion. The destination is warned about congestion. The receiver in this case adopts policies to prevent further congestion.
 - **Backward Signaling :** In backward signaling, the signal is sent in the opposite direction of the congestion.
- The source is warned about congestion and it needs to slow down.

- **Congestion control algorithms**
- **Leaky Bucket Algorithm**
- Let us consider an example to understand
- Imagine a bucket with a small hole in the bottom. No matter at what rate water enters the bucket, the outflow is at
- constant rate. When the bucket is full with water additional water entering spills over the sides and is lost.

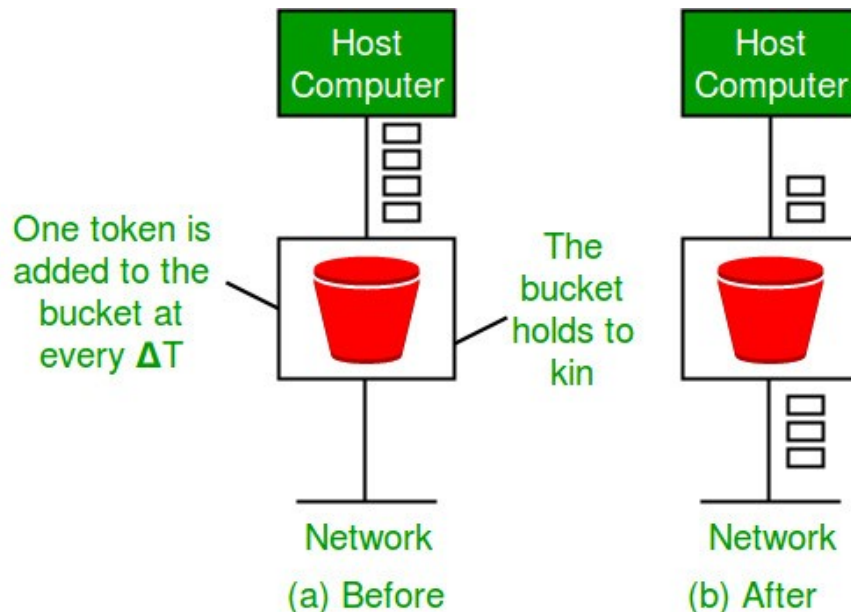


- Similarly, each network interface contains a leaky bucket and the following **steps** are involved in leaky bucket algorithm:
- When host wants to send packet, packet is thrown into the bucket.
- The bucket leaks at a constant rate, meaning the network interface transmits packets at a constant rate.
- Bursty traffic is converted to a uniform traffic by the leaky bucket.
- In practice the bucket is a finite queue that outputs at a finite rate.

- **Token bucket Algorithm**
- Need of token bucket Algorithm:-
- The leaky bucket algorithm enforces output pattern at the average rate, no matter how bursty the traffic is. So in order to deal with the bursty traffic we need a flexible algorithm so that the data is not lost. One such algorithm is token bucket algorithm.
- **Steps** of this algorithm can be described as follows:
- In regular intervals tokens are thrown into the bucket. f
- The bucket has a maximum capacity. If there is a ready packet, a token is removed from the bucket, and the packet is sent. If there is no token in the bucket, the packet cannot be sent.

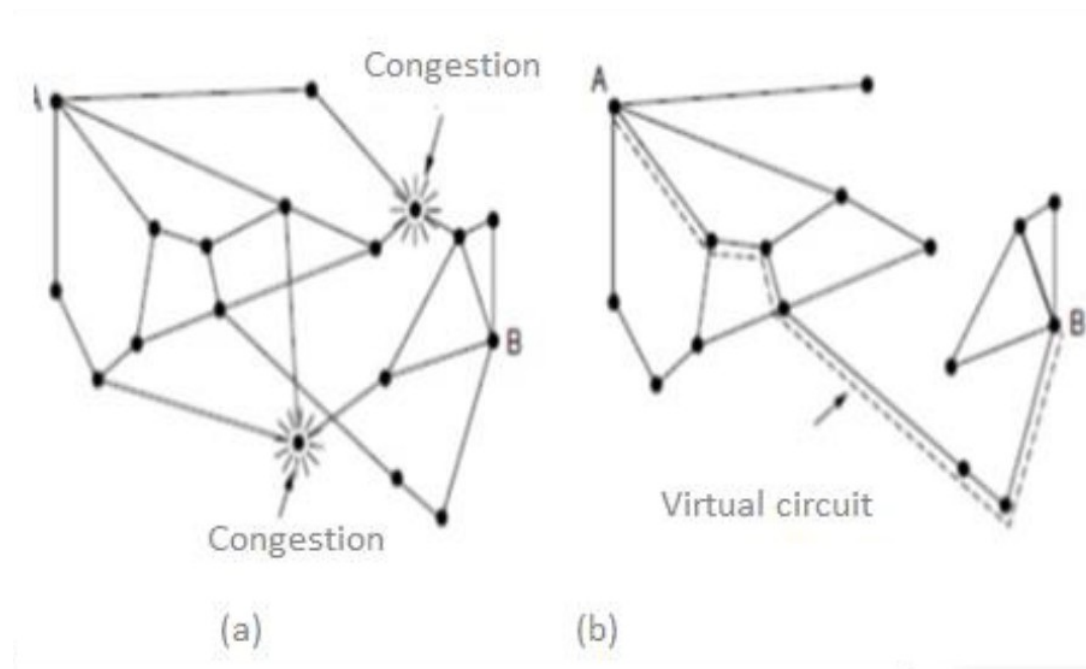
- Let's understand with an example,
- In figure (A) we see a bucket holding three tokens, with five packets waiting to be transmitted. For a packet to be transmitted, it must capture and destroy one token. In figure (B) We see that three of the five packets have gotten through, but the other two are stuck waiting for more tokens to be generated.
- **Ways in which token bucket is superior to leaky bucket:**
The leaky bucket algorithm controls the rate at which the packets are introduced in the network, but it is very conservative in nature. Some flexibility is introduced in the token bucket algorithm.
- In the token bucket, algorithm tokens are generated at each tick (up to a certain limit). For an incoming packet to be transmitted, it must capture a token and the transmission takes place at the same rate. Hence some of the busy packets are transmitted at the same rate if tokens are available and thus introduces some amount of flexibility in the system.

- Formula:** $M * s = C + \rho * s$
 where S – is time taken
 M – Maximum output rate
 ρ – Token arrival rate
 C – Capacity of the token bucket in byte
- Let's understand with an example,



- **Admission control**

- In computer networks is a mechanism used to prevent network congestion and ensure Quality of Service (QoS) by regulating the establishment of new connections or data flows. It operates by checking whether sufficient network resources are available to accommodate a new request without negatively impacting existing traffic or compromising QoS guarantees.



- Explanation
- **Step 1** – Suppose a host attached to router A wants to set up a connection to a host attached to router B. Normally this connection passes through one of the congested routers.
- **Step 2** – To avoid this situation, we can redraw the network as shown in figure (b), removing the congested routers and all of their lines.
- **Step 3** – The dashed line indicates a possible route for the virtual circuit that avoids the congested routers.

- Key aspects of admission control:
 - **Congestion Prevention:**
 - By limiting the number of active connections or the amount of data entering the network, admission control helps prevent routers and links from becoming overloaded, thus reducing packet loss, delay, and jitter.
 - **Resource Management:**
 - It involves assessing the availability of critical resources such as bandwidth, buffer space, and processing power before admitting new flows.
 - **QoS Enforcement:**
- For applications requiring specific QoS guarantees (e.g., real-time voice or video), admission control ensures that the necessary resources are reserved or available to meet those requirements.

- **Decision Parameters:**

Routers or network devices use predefined parameters, often called "flow specifications," to determine whether to accept or reject a new flow. These parameters might include required bandwidth, maximum delay tolerance, or packet loss limits.

- **Implementation Techniques:**

Various techniques can be employed, including:

- **Resource Reservation:** Explicitly reserving resources for specific flows (e.g., using protocols like RSVP).
- **Traffic Shaping/Policing:** Mechanisms like Leaky Bucket and Token Bucket can be used to control the rate at which data is injected into the network, indirectly influencing admission decisions.
- **Network Admission Control (NAC):** Involves authenticating users or devices before granting network access, often based on identity or security posture.